PW 1

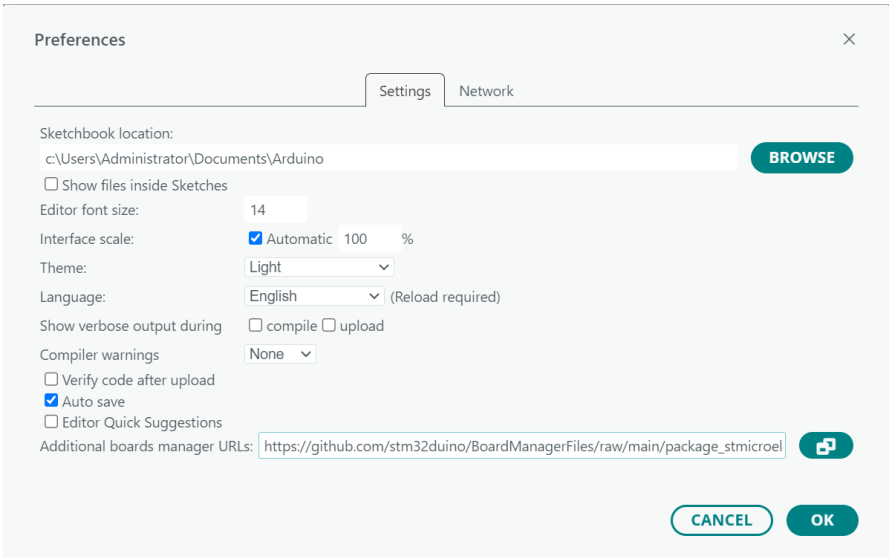
1 Set Up The Environment

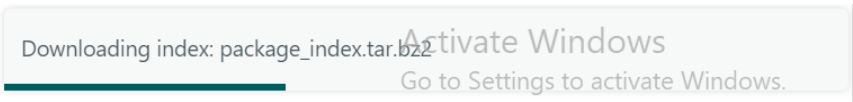
1.1 Install Arduino IDE

You can install it in the link Arduino IDE Link

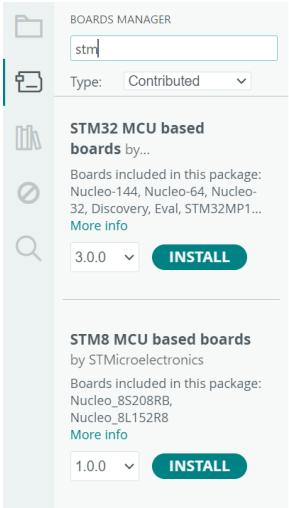
1.2 Install STM32 Boards

1. Open adruino IDE
2. Go To `File > Preferences`
3. Put the link `https://github.com/stm32duino/BoardManagerFiles/raw/main/package_stmicroelectronics_index.json` in the additional boards manager URLs
4. Click on OK which triggers an installation

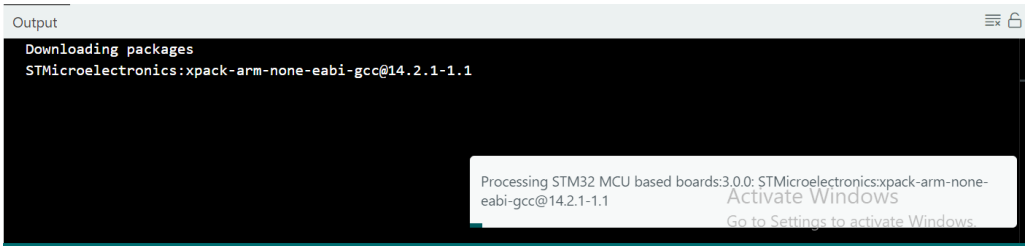


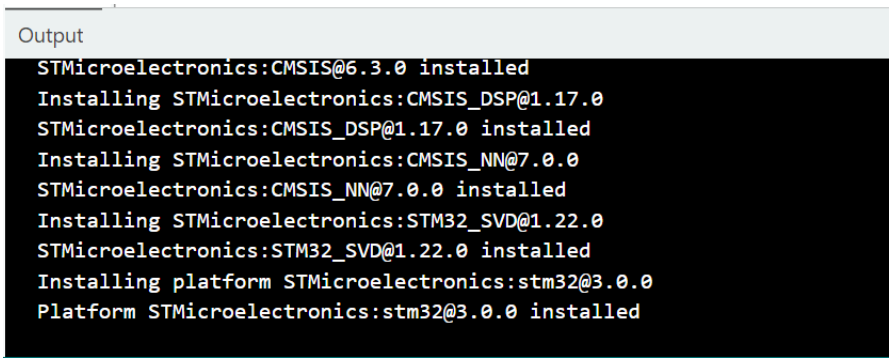


5. After the installation is completed go to `tools > boards > boards manager`
6. pannel will show to the left input stm in search bar and set type to contributed

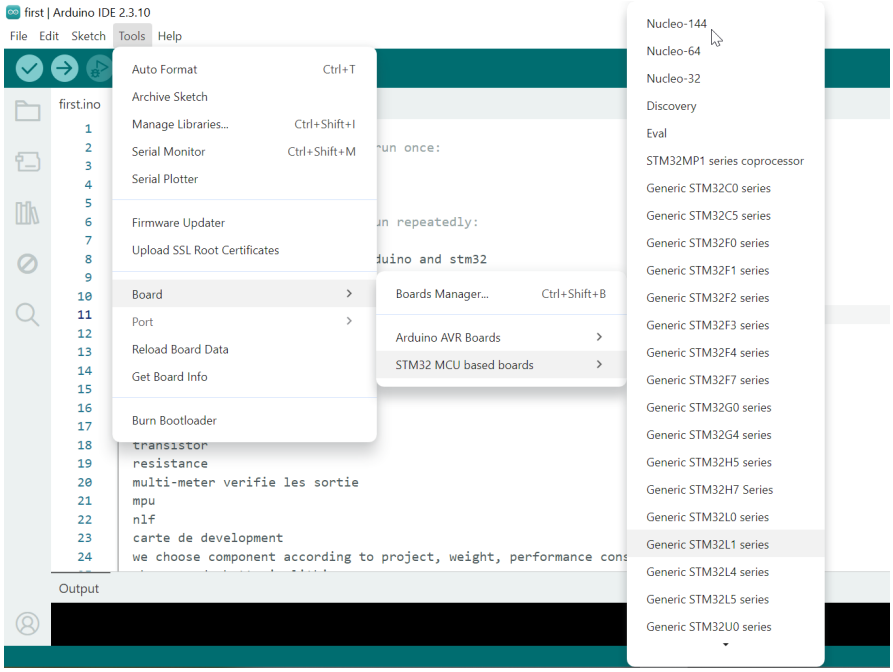


7. click on install for STM32 MCU bases boards, bottom panel will start downloading





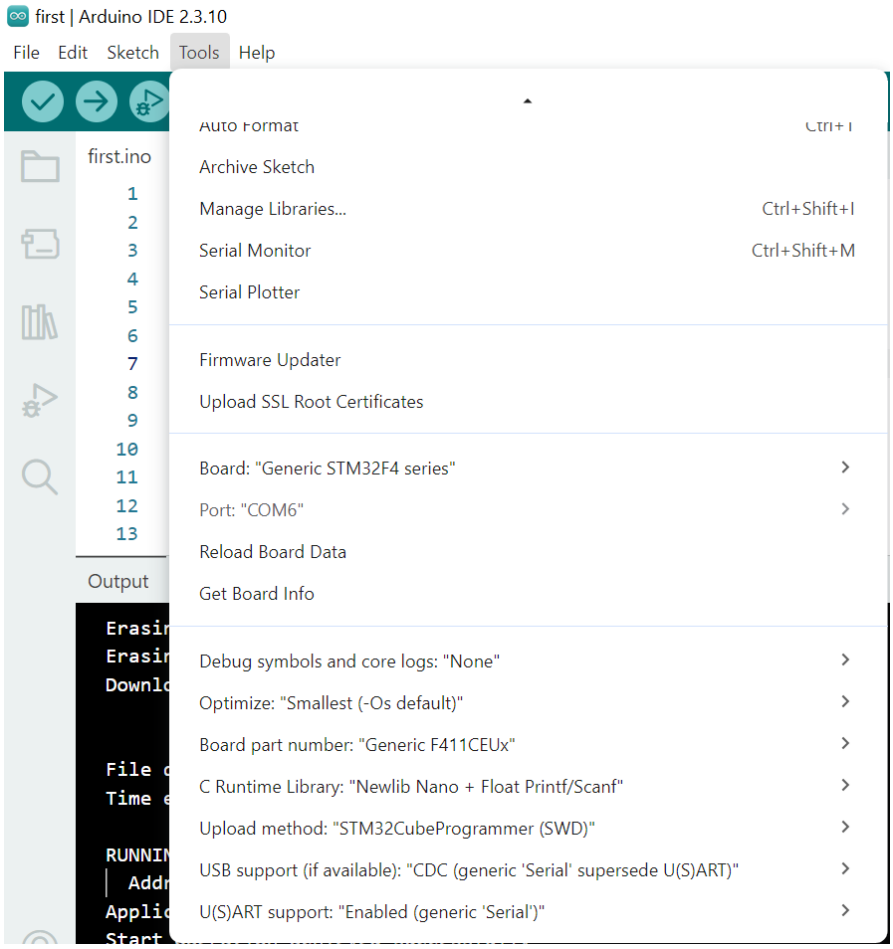
8. Finally to check if installed go to `tools > Board` and there should be `STM32 MCU Based Boards`



1.3 Install STM32 Cube Programmer

You can install it in the link STM32 Cube Programmer Link

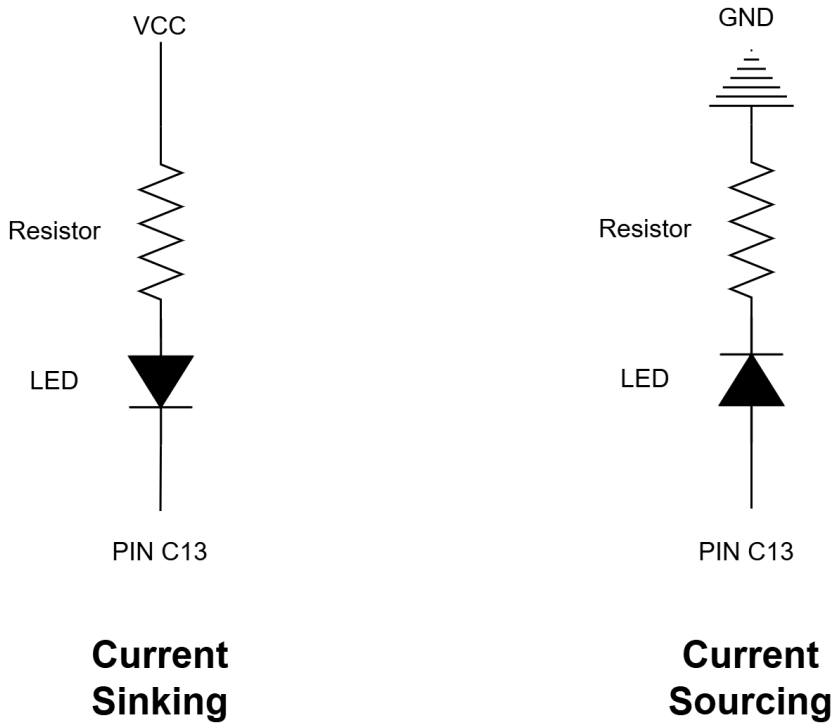
1.4 Set Tools Settings In Arduino IDE



2 Sinking & Sourcing Current

It is really important to understand this concept as it serves as a base to everything onward :

- **Sinking Current** : the active state is `LOW` because the resistor is linked to the **VCC**
- **Sourcing Current** : the active state is `HIGHT` because the resistor is linked to the **GND**



In our case the STM32F411 integrated LED uses a **Sinking Current** meaning to turn the LED On we need to put the pin to `LOW` and to turn it Off we set it to `HIGH`

3 Code For Turning On/Off LED On Pin C13

LED.ino

```
#define LED_PIN PC13 // define constant pin c13 of the LED

// setup function runs once
void setup() {
  pinMode(LED_PIN,OUTPUT); // set pin c13 to output
  Serial.begin(115200);   // serial monitor will send 115200 bits per second

  while (!Serial) {                // loop as long as serial not ready
    delay(500);                    // delay 500 ms
    Serial.println("STM32 INIT");  // print init message
  }
}

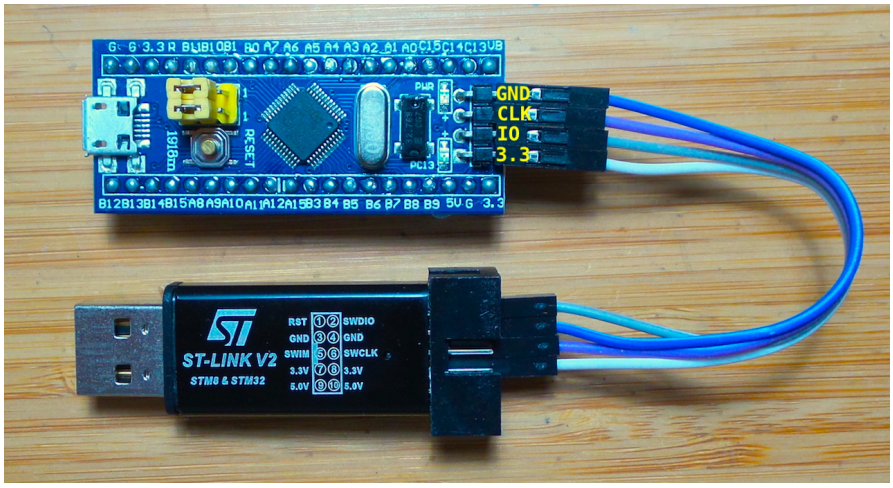
// loop function runs infinitely
void loop() {
  digitalWrite(LED_PIN,LOW);    // set pin c13 to low
  Serial.println("LED ON");     // print led on
  delay(500);                   // delay 500 ms

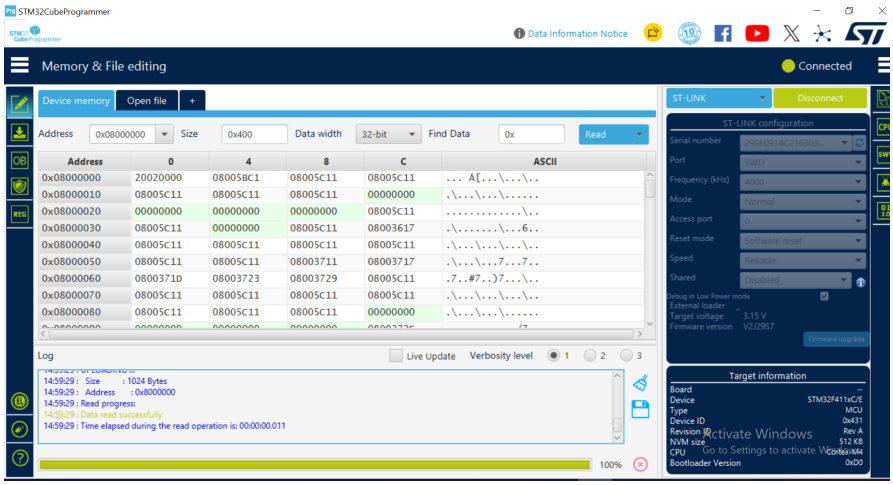
  digitalWrite(LED_PIN, HIGH);  // set pin c13 to high
  Serial.println("LED OFF");    // print led off
  delay(500);                   // delay 500 ms
}
```

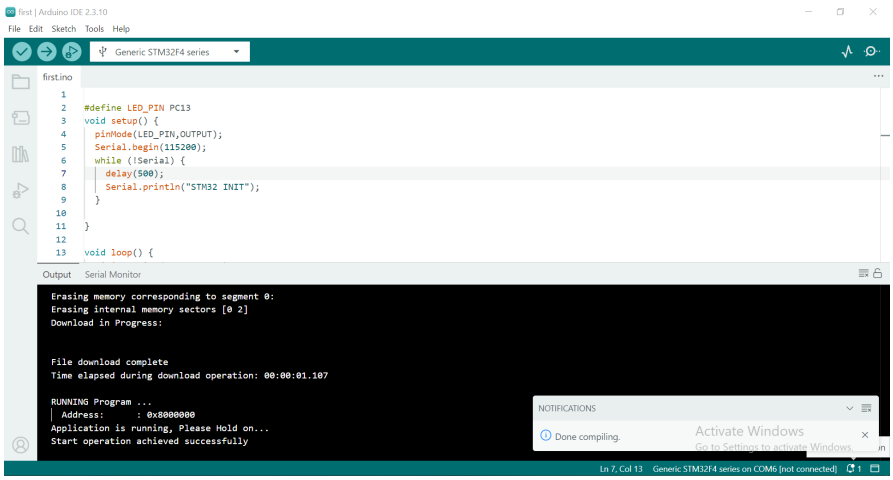
4 Upload Code With Stlink V2

We need to upload the code to the STM32F411 to do so we will use stlink v2 :

1. Link the stlink v2 with the STM32F411 using jumpers female to female , we need to link the four pins **GND**, **SWDIO**, **SWCLK**, **3.3V (VCC)** then plug the stlink in your PC
2. Test connectivity in STM32 Cube Programmer by clicking connect button for stlink
3. Click Upload button (icon right arrow) in arduino IDE

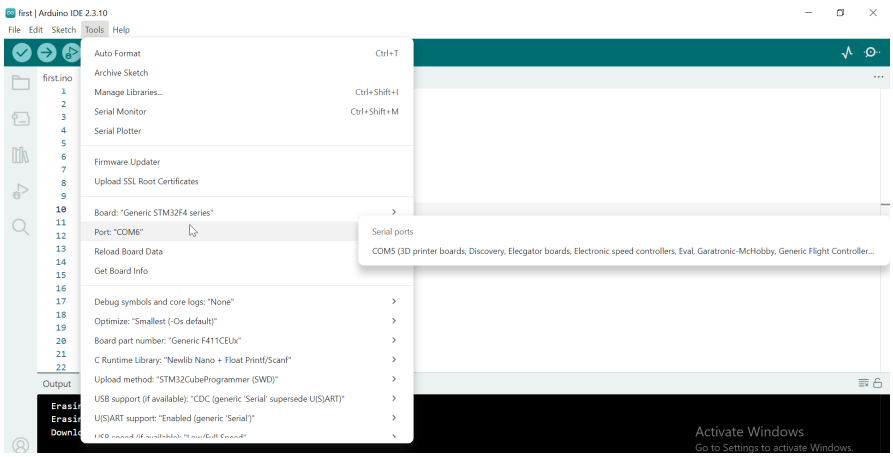


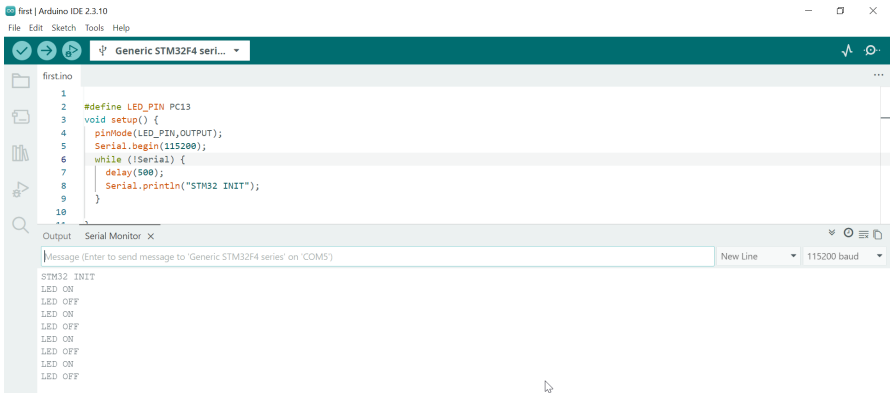




5 Execution

To execute the code use a type-C USB cable to provide power supply to the STM32F411 from our PC, then we will need to select a port for the serial monitor and set its baud (bit send per second) same as code in our case it is 115200



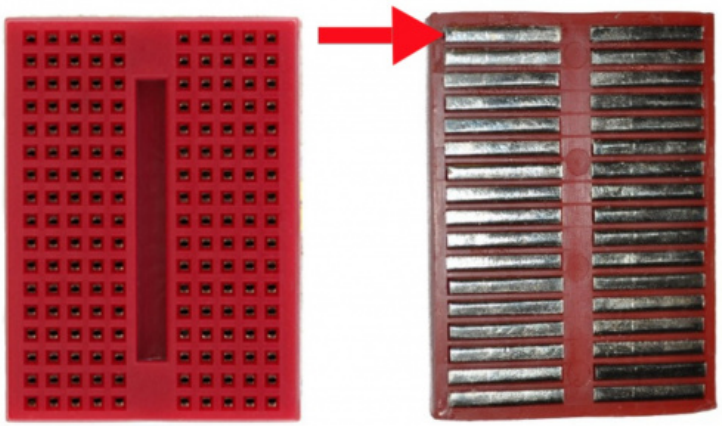


6 Breadboard

breadboard allows us to easily test and prototype our embedded system, in this section we will first explain how they work and then use it to create small circuit with LED and resistor :

6.1 How Does It Work

To first understand how it work we will need to see what's inside of it :

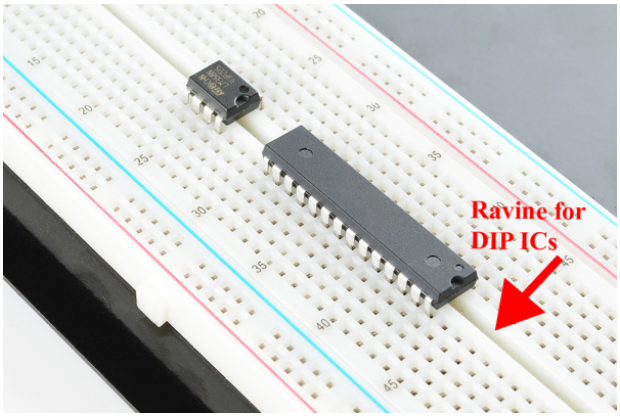


As the image show we can see that the breadboard is composed of multiple horizontal metal strips which allow connection between the holes of the same line.

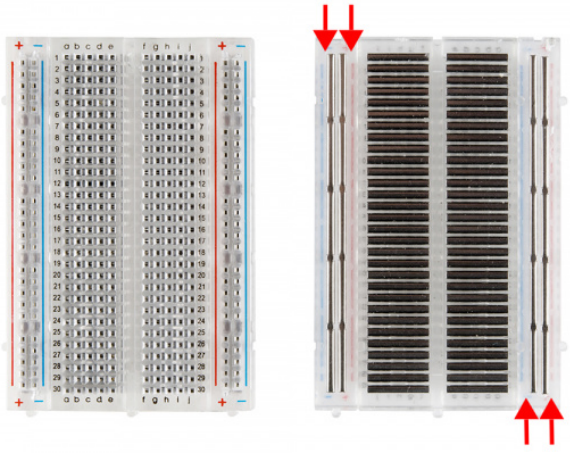


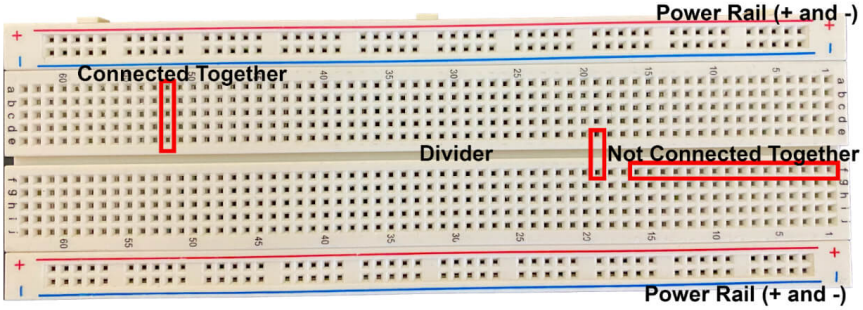
Each strip has 5 clips that's why each row has 5 holes

Note that the space in the middle is called a ravine (divider) It is there to ensure there is no connection between the left and right sides. It is deliberately designed to isolate them for DIP chips (like our microcontroller), because otherwise, different pins on the same row would be connected together, causing a short circuit.



On the outer edges, there are power rails each are isolated from the connection in the middle similarly, they have metal strips, but they run vertically.



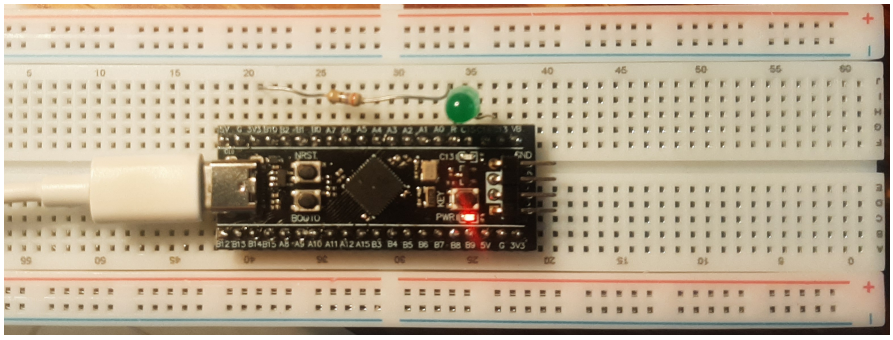


6.2 Practice

We first need to insert the STM32F411 into the breadboard, then we need to figure out how to connect the LED and resistor. To do so, we will try both methods: **sinking current & sourcing current**:

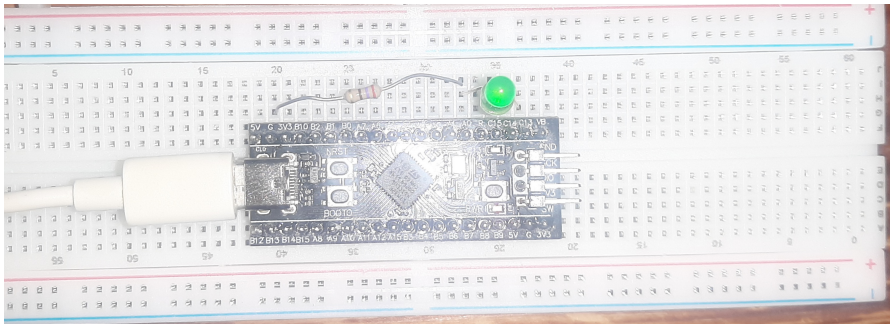
6.2.1 Sinking current

- we connect the first leg of the resistor in the same row as the 3.3V pin of the STM32F411
- the second leg of the resistor we can put anywhere, but preferably close to the pin C13 of the STM32F411
- the anode of the LED (bigger leg) needs to be connected on the same row as the second leg of the resistor
- the cathode of the LED (smaller leg) needs to be connected on the same row as the pin C13 of the STM32F411



6.2.2 Sourcing current

- we connect the first leg of the resistor in the same row as the GND pin of the STM32F411
- the second leg of the resistor we can put anywhere, but preferably close to the pin C13 of the STM32F411
- the cathode of the LED (smaller leg) needs to be connected on the same row as the second leg of the resistor
- the anode of the LED (bigger leg) needs to be connected on the same row as the pin C13 of the STM32F411



Note that when using sinking, the external LED and the integrated LED behave the same, whereas when using sourcing they are opposite.